ActInf Livestream #016.0: "Neural correlates of consciousness under the F

Livestream | 1:39:26

hello everyone welcome to the active
inference live stream
and this is the active inference lab i'm
daniel friedman
and i'm really excited to be having this
conversation with my friend blue
hi blue welcome to the active inference
lab everyone
we are an experiment in online team
communication
learning and practice related to active
inference you can find us on our
website twitter email discord youtube
whatever it may be in the future this is
a recorded and an archived live stream
so please provide us with feedback so
that we can improve on our work
all backgrounds and perspectives are
welcome here and we will
look forward to using good dialogue
video etiquette
today in 16.0 we're going to be setting
the stage for
february 16th and february 23rd 2021's
events which are both going to be
discussions on this same paper we're
talking about today
and for the 16th we're going to have the
author
um wanja whis join us for the discussion
today in 16.0 the goal is really to set
the context for 16.1
and 16.2 which are participatory group
discussions
the paper we're discussing is the neural

correlates of consciousness under the
free energy principle
from computational correlates to
computational explanation
by whis and carl fristen november 18th
2020
and there's a link given the video is
not uh
uh it is an introduction to the context
it's not a review or a final word
blue and i just gave it a read we're
still working through these issues we
hope to kind of
surface what we're thinking about on
those topics and one of the punch lines
is that fep it's not a singular theory
or definition of consciousness
itself but it lends itself to some
pretty powerful thinking and theory
development in this area
and in the video of 16 we're gonna go uh
with an introduction section and thanks
blue for the structural
tip we're gonna talk about the author's
perspective on
their introduction really aims and
claims abstract and roadmap
then we'll consider the keywords kind of
just where we were at with those topics
and where we think there might be a nice
bi-directional road kind of on and on on
and off ramp from active inference
and then we'll actually go through a
fair amount of the details of the paper
including some of the
formalisms close with a couple questions
and thoughts
and in 16.1 and 16.2 we'll be discussing

this paper
so that is all for the introduction
let's go to the aims and claims of the
paper
um the paper was as stated blue do you
want to um
read the aims and claims sure
the first aim is how can research on
neural correlates of consciousness
lead to an explanation of consciousness
and what role could the free energy
principle the fvp
play in this endeavor we have suggested
that research on neural correlates
should be
complemented by research on
computational correlates of
consciousness
cccs in order to yield a computational
explanation of consciousness
moreover according to the free energy
principle neuro
neural correlates of consciousness must
be defined in terms of neural dynamics
not neural states and compulational
correlates of consciousness
should be defined in terms of
probabilities encoded by neural states
all we hope to have shown is that a
computational explanation of
consciousness is possible
and that the free energy principle can
make a significant contribution to this
project
big aims and claims um and a lot of
interesting things
that are already being mentioned a lot
of these words we're gonna dive into

so if it's their first time hearing some
of these terms and acronyms
or whether these are things where you're
already bringing a lot of priors to the
table
hopefully you'll be able to look at it
in a new light and these are big topics
so
there's just going to be a lot
underlying these aims and claims
anything else you'd add on those um
cool let's talk about the abstract which
is how the authors have distilled and
summarized their work
for broad uh reading so this is the
section that most people
read it's kind of like a power law where
a few people see the title and the
smaller subsets see the abstract and it
goes down so we
always look at how the authors reflect
their work in the abstract
so i'll read the first half the abstract
and then you can give a thought on
something in there
how can the free energy principle
contribute to research on
neural correlates of consciousness and
to the scientific study of consciousness
more generally
under the free energy principle neural
correlates should be defined in terms of
neurodynamics
not neural states and should be
complemented by the research on
computational correlates of
consciousness
defined in terms of probabilities

encoded by neural states
we argue that these restrictions
brighten the prospects of a
computational explanation by addressing
two central problems
so just a thought there i think that um
the fep is really well suited to
providing a computational explanation of
consciousness or
a computational correlate of
consciousness um
if we can kind of maybe talk about
the easy problem of consciousness right
which which we'll get into the hard easy
and
medium problems of consciousness but
really the easy problem of consciousness
is is well
described by the fep and i don't know
that we can really even
in like nail down what exactly
consciousness is except
from that computational standpoint cool
and this question of defining
consciousness it's gonna
rear its head a million different ways
not the least of which when deciding
what to measure or which analogy
to use having a computational versus
some other type of adjective that you
could put there
so how about you go for recapitulating
that
last point which is there's two central
problems that are going to be addressed
and then continue
so we argue that these restrictions
brighten the prospect

of a computational explanation of
consciousness by addressing two central
problems
the first is to account for
consciousness in the absence of sensory
stimulation
and behavior the second is to allow for
the possibility of systems that
implement computations associated with
consciousness
without being conscious which requires
differentiating between computational
systems that merely simulate conscious
beings
and computational systems that are
conscious in and of themselves
given the notation of or the notion of
computation entailed by the free energy
principle
we will derive constraints on the
inscription of consciousness
in controversial cases for example in
the absence of sensory stimulation
and behavior we show that this also has
implications for what it means to be
as opposed to merely simulate a
conscious system
cool so the two key
problems the central problems one is
what happens in the absence of stimuli
and then the second big plot block of
text is
um the difference between actually being
conscious versus simulating it
and then by thinking about these two
central problems and using the fep
as our guide remember we're under
according to the title under the fep

not because of the fep or some other way
of talking about it
under the fep we're going to think about
how this notion of computation
is actually going to lead to some unique
predictions and
explanations and hypotheses that then we
can talk about in a really scientific
way
that will take it out of the land of
abstraction and well what about this
thought experiment and what about this
uncertainty here
and towards the realm of well this model
predicts that this kind of a brain or
this kind of an agent
will have this type of measurement or
response in this context
we want to have that conscious
consciousness a meter
the measuring device but without the
definition it's hard to have the
measuring device
and vice versa and so fep looks like a
promising way to sort of navigate that
paradox
let's go to the roadmap so this
is a paper structure with six
sections an introduction and a
conclusion and with four
internal sections the first internal
section and we're going to go through a
lot of these keywords in just a second
starts from neural correlates of
consciousness and moves the discussion
from thinking
neurally to thinking computationally so
that's the movement from neural

correlates to computational correlates
of consciousness
then the third section moves from the
computational correlates of
consciousness
into the free energy principle surprise
surprise
the fourth section goes back to
computational correlates
and moves the um the focus the regime of
attention
from computational chords alone which
have already been linked to the fep
to computational explanations of
consciousness and then the fifth section
brings together the computational
explanations of consciousness
with some of the first authors previous
research on minimal unifying models of
consciousness
and all of this is implicitly interwoven
with fep
which is kind of what is being tied
together with this paper
so any thoughts on the structure or the
roadmap blue
no it's good it was a good um flow i
thought yep a dense
paper so definitely okay to take it in
many
sections and skip around and skim it
through it first there's a lot of
ways that each person will read
different genres of paper
okay the keywords the part we've been
excited to
talk about i guess all of it though and
we just took the keywords

from the paper and laid them out
uh in an order and tried to present each
keyword in a way that was standalone on
just one or two slides
again thinking about the on and the off
ramp for active inference
so if you read this because fristen's
one of the authors or because it came up
in a keyword search for free energy
principle
but maybe you haven't thought about some
of these other topics that's one
mode of listening and then the other
mode of listening is like maybe with
more experience
or having thought more about these
topics where's free energy principle and
active inference playing into all that
and then beginner's mind and learning by
doing plays on both directions of the
road
so our first keyword uh well let's just
actually read through the keywords
uh free energy principle and active
inference consciousness
minimal unifying model neural correlates
of consciousness
computational correlates of
consciousness computational explanation
and islands of awareness
okay so first free energy principle so
how about you read it and i'll give a
comment on the first half and then we'll
flip
awesome so the free energy principle
provides an
information theoretic analysis of the
concept of existence of self-organizing

systems
it entails that self-organizing systems
at non-equilibrium steady state bound
the entropy of their sensory signals by
minimizing on average
a quantity known as variational free
energy as such
fvp is not a theory of consciousness do
you want me to read the second part or
do you want to
switch now um i'll give a thought and
then we can switch if there's two i
guess
but just that in the first paragraph if
you only
dipped a toe in you'd already find that
fep is not itself a theory of
consciousness
so the debate's not done we're looking
at what the fep
is that's the first part of the
paragraph it's a certain information
theoretic analysis
for thinking about self-organizing
systems for complex adaptive systems
and it's not itself a theory of
consciousness so i think that that's
just a really nice way to open
the tone of the paper and the discussion
so the second part here referring to
what is the free energy principle and
how is this going to relate to the paper
a fundamental constraint provided by fep
is that it only applies to active
systems that engage with their
environment
hence a computer simulation of a system
that minimizes variational free energy

may perform computations that are
also instantiated by a conscious agent
that interacts with its environment
but if these computations are realized
by a piece of hardware that cannot be
regarded as an agent
eg they cannot move or change external
states then they can at most lead to a
simulation of consciousness
i think that this is interesting because
it's like it doesn't really
um just like delineate what exactly is
an
active system and what what is engaging
with the environment
like for example if you have like an
unborn baby inside
of the mother is that like an active
system engaging with its environment
like can does it have any control over
like the environment of the environment
is delivered to it like all
non-stationary like self-organizing
systems that are maybe more passive
like i think about like ocean you know
creatures that just sit there like
i know that the fep does apply but it
just doesn't really delineate
exactly what those systems are yep
agreed and um so if this is your first
time
hearing about it's obviously a big topic
there's a lot of ways to go about it
also i had a question like whether other
physical systems like a proton are we
going to include that as an active
system but
there's a lot of um is that an active

you know it's engaging with the
environment yes but is it active
um so that's where we're going what is
simulation versus actual consciousness
where's the fep
let's in this keyword section think
about just what is the fep before we
go that far out with a paper because we
wanted to draw on the quotes from the
paper to see where they were going with
the fep
and now let's pull back to just thinking
about what is the free energy principle
and this is a slide that had been used
actually a couple
uh active streams ago but it
actually a lot of the terms got reused
focker plank
helmholtz non-equilibrium steady state
markov blankets active inference
everything here
except for some of these more um
specific like
sub process theories on the neural side
almost everything here was discussed in
this paper
so this is why it's so interesting to
think about convergences between the
papers because there's a lot of overlap
we're on the right track and we can
hopefully start to reduce our
uncertainty
just to run through these the focker
plank and this is from mel andrews's
paper
and the history of thermodynamics of the
fep so active 14 if you want to
listen more but the focker plank is

bringing in
a uh uh vector
way in the helmholtz as well of dealing
with vectors
and mathematics for certain kinds of
systems
namely these non-equilibrium steady
state densities which we're going to go
into in more detail so we're not going
to go into it here
and the free energy principle is a just
as we said on the previous slide it's an
information theoretic analysis
approach for these kinds of systems
doing interesting stuff
non-equilibrium steady state cybernetic
systems
and using that type of vector
maths on it and then it leads to all of
these consequence
falsifiable testable hypotheses the
process theories up at the top and there
could be other things as well
anything else to add on that blue not at
all
all right so to go into one of the
process theories that we
like active inference and this is from
another paper we've discussed
a while ago vassal at all a world unto
itself
and this was thinking about
communication so on
11 we said okay it's about active
systems engaging with their world
slide 12 we're going to think about that
in this kind of thermo info
statistical way and then 13 okay

let's bring in almost the real processes
and almost a systems engineering or just
a systems modeling perspective
to what kinds of systems are engaging
with the world active
with a generative model of their niche
and how are we going to think about that
under the free energy principle that's
where active inference comes into play
and this is a diagram we see in many
guises
where you have internal states and
external states which are separated
by an interface that can be modeled as a
markov blanket but we'll also go into
more detail on that later
where what goes inside or from one side
of the interface to the other
is sensation coming in that changes
internal states
internal states such as uh
different uh policy estimates for the
world or different policy estimates
for what one's self will engage in
results in different action patterns
which
influence external states which also
have their endogenous dynamics
so that's active inference that's what
active inference lab is like
thinking about and coming at for many
angles so that's one
little just um snapshot of active
inference what is active inference for
you
today blue or what does this make you
think about
so i i'm always interested in reflecting

on
the internal states like hunger
right how that results in an external
action
like a behavior like i'm gonna go find
some food
so i'm always interested in reflecting
on on this and whether it's
um it can be reduced to
computation right like we know if we
have a good estimate of what our
predictive model
is or our generative model is if we have
that
uh estimate then we know like what's
coming at us what is the input
can we always predict output like are we
just the sum of of
are we just a computation like i always
think about that so it brings that up
for me
yep this is a skeleton but then if it's
skeletons all the way down
where's the meat where is the actual
self or is it something
other than that maybe it's that that is
the cell for some other thing like that
so active inference active inference
eight we talked about it from a
reinforcement learning and a control
theory perspective we've talked about it
from a historical angle
with mel and from a philosophical angle
um in 15
we're reintroducing consciousness which
we also talked about
in nine with the projective
consciousness model

so that consciousness free energy
principle active inference discussion
was about geometry and the projections
and then the way that agents did active
inference on this projective geometry
we're actually not going to talk about
that projective
almost perry personal space relationship
with the fep
and this paper is going to take the same
topics go in a little different way
with more of a focus on the information
theory and on the generative model
and on the divergence of a few different
kinds of
surprise and measurements of the world
so it's another facet
let's talk about the main piece of the
paper which is consciousness
so how many ways to start or cover it
um just when we were planning this as a
lab and talking about it people
were just bringing up good stuff in live
discussion so someone said a felt sense
of is-ness
people were asking like where do we look
for consciousness
what do we measure and where what
spatial temporal scales what kinds of
systems
and many people just pointed out there's
so many
understandings here and uh areas of
study i mean
if one is curious there's a great wealth
of resources
um before we dive into each facet of
this

hexaflower blue what does consciousness
what like what interested you in this
paper what makes you curious to
study these things so i don't know
consciousness is one of those things
that's like
love like we don't really have a good
like definition for it we all kind of
like
think we know what it is right like we
think we know what it is and what it
isn't
but but it's it's like a very you know
ethereal
concept um so i don't know it's always
interesting to try to tackle these like
things that can't really be pinpointed
or like nailed to a certain
like board right like you can't really
exactly define consciousness
it's always fun to kind of take a
scientific approach to evaluate these
things right like reduce love to you
know oxytocin or whatever but it's more
than that like it's it's
i mean clearly consciousness is more
than what any one thing that like where
it's more than the fep it's more than
what we're able to
you know attach to it project onto it
but it's it's
always interesting to try to tackle
these hard problems and i love the
hexaflower
so i've got to say like daniel put this
together it's like a
super cool design that takes
consciousness as the center point

and then integrates all of these
keywords that we're going to be
discussing the minimum unifying model
islands of awareness neural correlates
of consciousness computational
correlation consciousness
computational explanation and then the
easy medium
and hard problems of consciousness right
so all these are surrounding the
consciousness is the center point i love
it
what was kind of fun about that and
almost meta is like i was trying to
arrange the list
it's like okay we got active inference
free energy principle keywords get them
out of the way
then we have consciousness related
keywords and consciousness itself
and so how can they be arranged linearly
and it was like there's no single linear
arrangement
everyone could have a different way of
seeing that
so then um it's just something that from
studying to bs
we see the hexagon and then i see it
more like when we want to reflect
tiling but a coherent tiling like
settler's a catan
you can play on a bigger board smaller
board but instead of just something
that's blocky it's like
it's yeah okay so let's go to the first
little
sub pedal because thinking about the
problems that

steer us or that motivate us in
consciousness research
so this is like not you know moral
proclamations on consciousness
this is about scientific study of
consciousness and about investigations
um about people who are coming together
from different perspectives not just
living from within um one specific
definition
so people are going to be kind of
playing fast and loose and having
different
takes that's kind of the wildness of the
field so there's
famous framing of the hard problem of
consciousness by chalmers
in the 90s and subsequent research
that's a pretty famous
um singular moment but
i just thought of you know easy medium
and hard problems but different people
are going to perceive
different difficulties for some people
they're really hung up on
mind and body other people it's like a
speed bump they just say
yeah i mean it's this other way so
that's just something that
studying the field of consciousness
studies is so interesting to me is
some people a word they just they spend
their whole life
investigating it other people they just
blow right by it or it seems like a
question that totally
it even if they're aware of it isn't
something super

relevant so which problems are easy
medium and hard it fits in that category
there are hard problems like
how could a physical thing or can of
physical things generate conscious
awareness or how we know there's a lot
of hard problems
um but it's an interesting topic what
did this make you think about
um so i mean i think like i already
divulged what i was thinking about in
the easy medium and hard problems of
consciousness and
how kind of the fep really lends itself
to like the easy
problem and like really like are we so
reducible to the easy problem of
consciousness like i mean we all kind of
know that there's something like
more than that and like when we get into
that more than that
like thing do we become then like
less scientific and more like spiritual
okay like
like do we take some leap of faith going
into the hard problem of consciousness
or
are we just like ascribing some property
like if we're gonna really be scientific
about it like consciousness is one of
those things that if you're going to
buy into it like it has to kind of be
everywhere like
i mean we make this like hard cut off
like if you look at research like only
higher vertebrates that have a central
nervous system those are the only
like things that are conscious in the

world right like and
some people have like lines within that
like only humans right
so really can we can we draw these like
hard
boundaries like i don't know i kind of
feel like if consciousness exists
it's pervasive but maybe pervasive only
among
living systems but then as a tree
conscious
so it's one of these it's hard it's a
hard problem
yes um agreed and the easy one
is almost the part that explains it's
like when people make fun of
daniel dennett's book consciousness
explained consciousness explained away
it's like yeah we know how a robot would
respond to a sensory
um you know cue to engage in action
but that's not the hard part that's not
really the
heart of the matter next topic so
minimum unifying model of consciousness
what
what did you find interesting about this
or what did you like about what you
read i liked this quote
that um i pulled out from from the end
of the
uh paper like i think it was right
before the conclusion maybe the end of
section four
that says you know minimizing
variational free energy could count as a
minimum
unifying model variational free energy

must be minimized by every
self-organizing system that persists
and hence also by any conscious system
i liked that and i thought that it was
applicable and kind of
really right on target
the challenge is decompiling
so every self-organizing system that
persists
are those the conscious systems are
those two sets the same so
self-organizing uh storm cloud what is
it like to be a storm cloud what is it
like to be a volcano
that's the question of what is it like
to be a bat and then
um another thread is like the cultural
like
maybe if there's a prior or something
that's implicit or
sub-personal that leads us to certain
kinds of
explanations being just absurd versus
intuitive like pan psychism versus only
vertebrates or only
subclasses or or nothing it's like what
do you mean nothing it's like well just
nothing it's not something i wonder
about
why not but not not an issue if not
so that's the minimum unifying model
it's going to come up in the terms of
this paper just because that's kind of
um both the authors are interested in
that
i would say and just to go
into one of the prior papers a solo
author paper by the first author

um longia so this is the science of
consciousness does not need another
theory it needs a minimum unifying model
unifying words on 7 11 2020
great date for pape and here
are on the right side are referring to
two kind of hypotheses for how we could
think about different
fields of meaning one way of
uh arranging different fields would be
like information generation is
all these tan things in a so this is
like computers and random number
generators and die
or something like that and then
consciousness is a subset where maybe
there's some conscious systems that
don't generate information maybe there's
some that do
maybe you have a different feeling about
how these are overlapped like in b
where there's information generating
systems that's a big super class
and then within that there are systems
like the blue the non-reflexive behavior
that are
cybernetic to some extent like it's
predictive it's acting in a way that
might have
some game theory like or some sort of
adaptive seeming dynamics
but consciousness is a subset of that
and so i just really
appreciate the the research direction of
the author
because um the difference between a and
b is
is night and day and there's many other

options it wasn't an attempt to be
exhausted it's like
if we could talk about this we could
have a better way of framing oh our
computer's conscious our human and the
computer what's conscious
let's say let's take a step back are we
where who
what are we overlapping with what and
what's on the table here and what kinds
of words do we care about and where's
the territory it's okay if territories
are nested within each other or
overlapping
or they're intercalated or they're
they're complementary or something
but we have to pull back to kind of a
level where we can at least sort through
all these theories of the science of
consciousness
because otherwise it's just a total mess
and the information overload aside
it doesn't even make sense just if
everyone's working in their own
wavelength it doesn't even make sense
so really nice um direction any thoughts
on that blue
so i wonder if like information
generation and consciousness like aren't
a full overlap
like if those can just like be all
lumped into one of one another and then
non-reflexive behavior is like some
subset
of that um just just even going to like
the spin of a of an electron like a
coupled electron like if you're looking
at

at qubits right like so it it's always
got to be there's a paired system and is
that conscious
like just getting down to the very basic
like atomic particles
especially what we talked about earlier
like if the free energy principle
is for so-called passive and active
systems that are engaging
with their niche like systems that
aren't usually thought of as very
reactive like
a proton or like a rock at a larger
scale
are they on this consciousness continuum
or are they disjoint from it or
so those are the kinds of things where
reading the paper
helps bring us to the edge of the ideas
and of research and also to the edges
socially thinking about it and talking
about it hearing people who have
different
perspectives so on the perspectives
issue also
computational explanations so just in
the word
what is computation you're going to have
a lot of takes some people will say
i mean of course you know my linux
computer right here is a computer
but then they'll say that the hydrogen
is also a computer and the football when
it gets kicked it's also computing a
parabola so it's
computing everything and then what is
explanation
philosophy of science rabbit hole to go

down
but um is it about being emotionally
satisfied with what we know
or is it about being able to usefully
act like do we have an
explanation for traffic i don't know
maybe some people would say yes no in
this situation yes but
do we have best practices do we have
approaches
so explanation versus approaches and
then the last point here
is just thinking especially in the
biological case about
unconventional computation so let's just
call the turing
von neumann genre the binary state
machine
idea conventional computation
there's a whole world quite literally of
unconventional
computation so reservoir computing
things that we're going to talk about
right at the very end
within the binary computers and then
outside of the binary computers physical
computing
wetware by bray is kind of about moving
beyond software hardware
distinctions which don't exist in cells
dna is dna it's not hardware it's not
software it's not source code it's dna
and then leviathan the cover of that
famous work
with a larger the group
made up of the individuals so that kind
of multi-level systems perspective
on computation multi-level cognition all

of these things come into play when
we're talking about
computational explanation
so can i just like maybe add a little
bit
oh yeah i know because i know that this
is one of your chillest areas
so i just think about explanation like
what is the
um like explanation is like the death of
curiosity
it's like the ultimate like pacification
like i just explained it to you
therefore
shut up like that's why that's your
answer so i think like
explanation like we should never be
satisfied with any explanation
like we should not presume that we
understand fully
like anything because it just ceases
investigation at that point
um and so so i think like you know
computational
approximation might be more fun and
and lend itself more to deeper
scientific explanation
yeah one thing that we were talking
about just like um an hour ago
with some other people in the lab was
like if the heart
of the issue is how things are realism
once you get the lossless
you know mp3 file the quest for what is
is done
but then if the song is not just the mp3
file but it's actually the relationship
how could you say that that ever ends so

it's kind of like
yeah are we looking to engage and to do
things
in the world and act or are we looking
to come to
a satisfying emotional explanation in
some
uh topic all right so
19. so here we are getting to some of
the
details like the computational versus
the neural correlates of consciousness
so there's two double c's this is the
ccc and then the next one is going to be
the ncc
so what are the computational correlates
of consciousness
what would be your definition
for ccc so
like i said i think that the
computational correlates of
consciousness um
you know it goes back to like is a
football conscious right like it's input
output
so like you know you have you what what
goes into the system and what comes out
of the system
and and you know is like can we call
that consciousness
i don't really know so but but for me
like my interpretation is
is it's a decision right like so you
have the input you know what you're
supposed to do or what you think you're
supposed to do you have some
approximation
about what you're supposed to do how

you're supposed to engage with your
environment based on the input that
comes in
and then you act in accordance with what
you think you're supposed to do or
or what you predict will lead to the
best outcome
i don't know something like that yep
well
the correlate word is suggesting that
this is
not an explanation that's causal
correlation is not causation
as they say so it's like we're talking
about correlations we're talking about
observables
kind of like a correlation like a
regression coefficient
but a little bit probably deeper than
that and then
i guess my question for the authors
would be how is
this related to the cybernetic
correlates of consciousness or the cyber
cybernetic attributes um in the paper
nine with rudroff and
um and other authors it was like there
was the five
functional invariants or something like
that like the perceptual invariance
so what are these computational
correlates um
and how are they related to algorithmic
or cybernetic
because maybe the word computer has a
lot of baggage we're live streaming on a
computer it's just
used in a really specific way it's a

specific word and then
maybe there's another way to think about
like processural correlates or like the
informational correlates of
consciousness that might be a little bit
um different aspect of what they're
showing but the details of what they're
showing we'll get to
all right ncc so the neural coral is a
consciousness it's another
correlate question so we're looking for
neural things that correlate with
consciousness so first off correlation
causation again
and the correlation can be like
evolutionary like we think that this
brain region over the breadth of species
diversity is correlated with
consciousness or it's correlated with
this behavior
or the correlation can be like a
molecular thing
or it can be a dynamical pattern this
fmri pattern or this eeg pattern
is a correlative consciousness we set
them up so that it was an experiment
where in
this task they did have consciousness
and this one they didn't
so it gets into defining consciousness
but they say the difference between
these two conditions is the neural
correlated consciousness
or the difference between patients in
this category versus patients in this
category is this
and for two takes on the issue um in
2016

um christoph koch at all um talked about
like
from their pro neural correlate of
consciousness agenda
direction what were the problems and
progress
in their fields and then here's a paper
from 2010 2019 with me
and a colleague where we kind of took a
step back from just thinking about the
neural correlates
and thought okay what about for the ant
colony or for other systems that aren't
just
like set up like the mammalian brain
how would we think about consciousness
because if you study only the neural
correlates consciousness
you're studying how would you know that
that's actually the subset of systems
it's like you're shining a spotlight
maybe a good one maybe a distorted one
on the neural system
but what if it's not just neural it's
glial oh well of course i meant that
okay
what if it's neural and the body oh yeah
i meant that too so all of a sudden
okay now you're talking about neural in
the niche and that's why we're talking
about multi-scale systems
so that's what there's a lot of
interesting stuff happening here and
about how the fep and multi-scale
approaches
fit together with the more neural
focus and the more certain time scales
and certain neural signatures

that avenue of research in neuroscience
humans
sensu strictdo now connecting it to a
clearly very vast area what do you think
about that blue
um so i think that the neural
correlative consciousness is cool i
think that
um like they do the fmri study like when
you're asleep versus when you're awake
and then they take like the difference
that's like a common way that they like
estimate
the neural covalets of consciousness
um there's also like the hyper
consciousness like
um who's i can't remember the author
um but he did like the information
theory like aspect of like psychedelics
so like expanded consciousness like what
does expanded consciousness look like
and how like there's
more connections between uh neurons i'll
have to remember the name of that paper
for next time
um but but it's it's just uh the neural
correlation of
of consciousness was really done by
differentials
that's a common thing i think yes
exactly it's an experimental way to get
at
using measurements real data real
experimental setups the kinds of things
that exist
in neuroscience labs getting at
questions of consciousness
and the question is are we are we on the

right track or might be misled by the
tools we have available and approaches
we have
so 21 islands of awareness okay so this
is kind of a topic that i'd never
heard about it's from a recent paper by
bain
at all 2020 and so we're thinking about
kind of the fringe cases or the areas
where we might be able to have a unique
prediction or an explanation
that is going to be a value add for the
fep
in thinking about neuroscience and
consciousness or which systems are going
to be useful to think about
uh even if other systems also agree and
there's going to be this case proposed
like a thought experiment called the
island of awareness
which is a conscious stream so i don't
know is the rabbit already in the hat
it's a conscious stream or system whose
contents are not shaped by sensory input
from either the external world or the
body and which cannot be expressed via
motor output
now one could just say well but what if
the body's part of the whole conscious
system so it's
it's a kind of non-sequitur to say well
it's happening in the brain ergo you can
cut that out
maybe you can maybe you can't but by
looking at the paper
and by believing that the authors have
thought through what they wrote like
it's always a useful starting point for

discussion
so the fringe and the edge cases can be
where we start to see theory
differentiation
so if this thought experiment isn't very
interesting to you you'll probably find
another one that
might be but these are like we're
looking for the exceptions that prove
the rule
or the um sort of higher level
game that is revealed by us over
sampling
just one area of state space the issue
with thought experiments though is
well kind of like science fiction they
can provide inspiration if they're
um at the right place the right time
you're never gonna reach these ideal
situations
and they often have a huge amount of
assumptions built in and then even for
uh there's this dream of empirical
application oh we'll study the island of
awareness then
because it's such a great thought system
and there's a hundred papers on publish
on it
but then it even for a close case like
oh
maybe in this medical condition there
locked-in syndrome
it ends up that it's not exactly what
people want and so then
it goes into a really protracted
multi-year thing with just
is the system the right one to test this
thought experiment short answer

no and it's hard to know what to do with
real systems to really make a dent in
the consciousness problem
any thoughts on that um
just that uh you know the islands of
awareness i think
um are always there's always going to be
those exceptions to the rule or those
hard cases like we don't even really
know how to
get at consciousness in a system like
that like a brain and a dish model
is there consciousness in that like i
remember working in the lab and growing
cells
in my like petri dish or whatever they
were growing in and like
you know i i was like i wonder what my
culture is thinking or if they're
thinking like
i would be like doing neuronal
differentiation and be like you know
i just wonder like what they're thinking
about today so so i mean it's just
it's always like this mystery so we
don't really know if there's if we can
approximate consciousness in a system
like that also uh the author
of the information theory and
psychedelic paper is andrew gallimore
is his name sorry cool yeah it's um
and again like how does philosophical
uncertainty
or debate propagate through basic
research and application
um what about with organoid research on
neural cultures and so
is there a certain size of brain or what

about there's a brain in a body or you
recapitulated something from a
stem cell from a neuron and from a you
know skin cell or you made some new cell
type it's like
is it different or worse or better to
throw a gallon of skin cells versus a
gallon of this type of cell
those are big questions and that's why
the consciousness discussion is really
important
to not just resolve but to sort of like
clarify and to engage with so the big
picture
of this paper now to get into the
quotes and some of the key topics we're
not going to go through every quote
we're not going to read every quote
because there's
a bunch of quotes but this section
is going to just describe the main
three observations that link three of
the big
tlas three-letter acronyms together so
fep is the big one
but then we have the neural and the
computational correlates of
consciousness
so they're all kind of gonna be joined
together
and just to read the first point of each
one
if we're going to go fep neural
correlates should be in terms of
neurodynamics not neural states
so in other words patterns of neural
systems through space and time
not just like up the jennifer aniston

neuron fired
so they were conscious of jennifer
aniston it's going to have to move a
little bit beyond that
into something that's about dynamics
through state spaces
this second point is that there are
relevant distinctions to be made
between the probabilities of movements
through this neural state space
and the probabilities encoded by neural
states
that's going into a whole encoding
representation question
and a lot of statistical details if that
sounds interesting
read it and then the third point is that
some
of the computational correlates
literature
are able to be recast
within the fvp and so the fep is going
to be like a bridge
or like a nexus between people have
identified 10 computational correlates
and 10 neural correlates well you got
this blood signal in the
this brain region when they see this
thing in this auditory region when they
hear this thing and
this when they remember themselves and
then on the computational side you have
this feature
a b c d all the stuff like information
theory how are we going to bring them
together fep
any thoughts blue all right no
so let's go through kind of with a

little i don't think we kept the exact
order of the paper
but early in the paper we start from the
neural correlates of consciousness which
is something that
people have at least probably thought of
like there's something different when
there's
blood loss to the brain than the hand so
people have kind of thought that there's
a neural correlate
of consciousness we're going to move the
discussion from just thinking about
neural correlates to computational
correlates so this is the
narrative trajectory from being like
neuroreductionist or centric
to more computational perspective they
say ccc
computational correlates are more
general and promise to be more
explanatory than neural
correlates but there's also challenges
so big benefits that you get you can
consider ant colonies and ecosystems all
kinds of cool stuff and
perhaps have access to a whole new set
of computational tools
downside are what
computational correlates are not defined
in terms of neural structures
but are neutral with respect to the
question of whether the conscious
experience correlates with global neural
activity or not
so the issue is that the brain is
changing state all at once
and so you don't know whether that one

feature you isolated is really the
difference that makes a difference
so to speak it's like you could run two
different computers and you wouldn't
really know
what about their outcomes from a
correlated perspective
was the actual explanation so one part
is it's hard to get
an explanation even though they say it's
more explanatory
um and then basically
the second challenge for the
computational correlates
is the mapping from any correlate to
consciousness
especially in controversial cases and
the third challenge for any correlate of
consciousness
is basically fringe cases and cases we
don't know or don't
but even those who knows what we do and
don't know so
there's these are three of the major
dimensions
of the challenge for any correlate of
consciousness
global versus local structure
mapping between um consciousness and any
correlate
arbitrarily or not and then cases we
don't know or understand
so there's big territory that's
pretty open and there's a lot there what
do you think about that
there's a lot there's a lot of remains
to be you know elucidated
in all of those um areas so

yep our problem yep so 24. so yeah
kind of taking computational just
at the author's word that's how that's
the thread they tie
so it's a broader understanding than
just i think a computer processor i hope
that we'll
find that out especially when we talk to
the authors so there's a generality of
these computational principles but they
also have a disadvantage
the properties described by them may not
be necessary for many types of conscious
experience but probably not sufficient
so probably it's a probabilistic
statement but do we know what is
sufficient
and so in other words what is required
for consciousness that's sort of the
issue with this computational approach
is
you get this general toolkit that can
cross scales really easily say wait if i
can make a measurement of a millimeter
in a measurement of kilometers
and then i can use um my tools on
numbers
then it's a multi-scale model right and
so the challenge is
have you underspecified aspects of the
system
leading to a really um potentially uh
like bad false positive or false
negative rate
of course the question is what is the
ground truth who's ground truth
and how are we going to get to these
edge cases and resolve them in a real

finite world
and then the authors say in order to
address this question
of what is necessary and sufficient we
show that
we show to what extent and from what
assumptions some computational
principles can be derived from first
principles
so they're going to say all right this
even the computational correlates of
consciousness people yes there's some
issues with going computationalists but
we have studied those this is good
phrasing
then we're going to ask how even the
computational correlates could be
themselves generated from an underlying
something first first principles which
are going to be information
and physical principles as we are seeing
what do you think about that so i mean i
think
in order to delineate what's required
for consciousness you need to kind of
get at the point of what is
consciousness like and i think that you
know
they artfully skate past that in the
paper and i mean i don't blame them
right like
it's a it's a touchy um and
difficult and many gray areas you know
exist within trying to define
consciousness but if you can't define
what is consciousness how can you
define the requirements for
consciousness yep

agreed and so let's uh see what any of
the authors have to say on that but
this is the question how is
computational principles or physical
principles going to come into play
all right here we go into the formalism
section there's a bunch of formalisms in
the paper
and um here is just one from the paper
starting on the top left
that sites out to a fristen paper this
first in 2019 long monograph that is
having many
active uh improvements and revisions
and discussions on it and then here's
two sections from
this first in 2019 paper from 12 and 13
i think
but really you could take it even back
further just like any other literature
you kind of take it back as far as you
want to learn
but for the purposes of the top left the
paper that we're
reading today the key formalism
is here or one of the key formulas just
the one on this slide
we're going to express the flow of f
in terms of surprisal and hence in terms
of a non-equilibrium steady state
density
expressing the flow f in terms of
surprisal is almost like
thinking about surprisal like it were a
flow
so just like you could have a flow of
water we're thinking about
kind of surprisal flows which is to say

using flow
like dynamics to study surprisal because
that's
all sort of a few different facets of
the same thing
and so there's going to be some flow
that's going to reflect
some terms that can be rewritten a few
different ways we're not going to go
down all the details i think of every
letter anything that we should add on
this slide
blue but this is just about the idea of
expressing information as a flow
which is going to set us up for some of
these later discussions and formalisms
yeah no i'm good yep and we just throw
these slides
uh in the order just that we
snap them so there's other orders and
other
ways of going through it we had to look
up a ton of stuff we learned that the uh
i'm gonna bring this on the dell or the
nabla
it's the operator used in mathematics
particularly vector calculus
that is a vector differential so it's
like div
grad curl but it's it's an upside down
triangle but it does have to do with
change
just like a regular delta but it's a
whole area
of math and so this is just to say some
people
maybe they didn't expect an equation in
a consciousness paper

other people maybe they're very familiar
with this math
and applying it to consciousness seems
like an outrageous
overstep so that's the conversation
yeah if you're somewhere in the middle
like
i think i'm i'm somewhere in the middle
of that um but but if you're somewhere
in the middle
just knowing that it's like about
information geometries like how daniel's
mentioning it's about a flow and so
so when you have like it's it's a
geometry like the probabilities are
thought about in these shape
spaces and kind of knowing that helps to
contextualize
uh the way that it moves i think you
know thinking of float it all makes me
think of like
a program it's all set up and loaded
which files to read in what to analyze
you hit enter
it's like a bunch of grain in a silo and
you pull out
the stopper and then it flows because
there are forces so the computer is
having energy course through it
and there's certain irreversible
processes that
increment a flow of information
and so there's that's kind of i wonder
what kinds of metaphors like this could
be something for the author like
how do we think about this flow and
where how how do we what how can we
describe it and learn about it

so this is another this is so yeah that
maybe
this one could have gone before the
other one but
this is um a pretty general framing
of random dynamical systems that are
using this
langvain formalism that was developed in
the direction
of free energy principle by
ao 2008 and other authors
and basically we can read this bottom
part go back to the equation so the
state of the system
x of t at time t tau
is constituted by slowly changing
macroscopic variables
which are grounded in microscopic
variables with faster dynamics
so it's almost like saying little things
are composed
of littler things little things compose
bigger things so it's kind of
allowing us to go multi-scale by saying
that there's going to be a sort of big
wave and a ripple
and a bunch of the ripples summating
makes the big wave
informationally just kind of like a
bunch of physical things
spatially compose a bigger thing but
this is happening
through time as a random dynamical
system
not with a spatial metaphor so think
about little things assembling into
bigger things
and then pour it out of the physical

into the dynamical systems
and it's what kind of a dynamical system
it could be related to information
this is why the result is a stochastic
differential equation
comprising not only the state-dependent
flow f
but also a stochastic term omega with a
mean equal to zero
so another way of thinking about this is
like signal and noise
and so it's all like the vibration but
you can partition it
with a model into change through time
of a signal and a noise that's like much
higher frequency
so that noise it's not like you're
saying it's like bad it's just that
with the signal that you're pulling out
it's the residual
and the signals that are hard to find
are the ones that are shorter like
they're hiding in the noise
but those can be pulled out if you have
a really long time series you can
totally pull it out
and there's other signals where the
signal is so strong and then the noise
is so tiny like the vibration
an inch off of the orbit of a planet
it's like really easy to pull out the
signal
because the noise term is so small so
any thoughts on that blue or we can
continue all right
so now we want to talk about what exists
which is related to fristen's theory of
every

quote thing or of um particular physics
which is
a specific way of doing physics and it's
a physics of particles
anything that exists will have
characteristic features that do not
change over time
this is like measured theory it's like
if it's
not there through time it doesn't exist
like the reason why there isn't it's
kind of like
um material costs like if there isn't a
cow
in my room it's just not there something
doesn't persist
it's not there formally we express this
by assuming
that the system is a random dynamical
system in non-equilibrium steady state
so the equilibrium the total heat death
equilibrium steady state of the air
molecules in the room
would be dispersed to homogeneity with
some
summary statistic like the well
mixedness but if it's not
at equilibrium to the well-mixedness
physically
or kind of informationally then it's an
in a non-equilibrium steady state
the steady-state part is because of
existing through time
like the length of your arm but then the
non-equilibrium part is reflecting that
the arm is an
organization beyond just dissipating
air molecules so when you have a

non-equilibrium steady state like an arm
that's persisting despite being
disordered surrounding
this means that the system is exchanging
with its environment and its dynamics
can be described in terms of its nest's
probability density
i.e a probability density that does not
change during the time over which the
system is said to exist
so over the second to the second time
scale there's a constant measurement on
the length of the arm and through time
it could be modeled as something
changing
but in all those cases it's like
definitely not just
um something that's there's something
different about what that system is
doing
informationally relative to other
systems
okay and then the third kind of quote
section here is
such a system has a random attractor
which could be thought of in two ways or
another way to say that would be you can
think about systems as having a random
attractor which then has multiple
downstream consequences
first it can be considered as a
trajectory of systemic states that
evolve through time
so then in this sense the system will
after sufficient time revisit particular
regions of a state space
the attracting set so we've talked on
the active stream before about like

returning to a desired body temperature
so it's not that your body returns to
the state that it was as a body
a week ago a year ago 10 years ago it's
that
on that one dimension with respect to
the modeling of the signal and noise of
body temperature through time
because you're staying warmer than the
background for a persistent amount of
time
in a way that defies the otherwise uh
dis
dissolving forces of uh thermodynamics
you can model that like it's doing some
sort of strange attractor
that's what that part is saying and then
the second part
is that you can think about that as not
just you can take a trajectory
view like a single path through stat
state space temperature going up in the
morning or down in the morning
but the other way is like when you have
a million paths in infinite number of
paths
it's like a density and in fact it's a
flow density so it's like
one traffic you're driving that's like
one car
but then there's the traffic flow and so
when you switch from a particle
modeling to a flow modeling is
there's a lot of it's kind of like
discrete versus continuous time
but it's a spatial analog whether you go
particular like an electron as a
particle electron is a wave

how do you make that distinction and so
that these are all kind of related
topics but that's what is under the
scope of
the pulling back to thinking this way
anything else all right 27
the fact that internal states are
conditionally independent of external
states
given blanket states allows us to
disentangle systems dynamics so now
we're going to talk about the interfaces
between those active interesting systems
and their otherwise disordering
environments relatively speaking
or whatever their other side is whatever
the other side of the interface is
um in other words we can write down
separate equations of motions for the
internal states and active states
that only depend on the blanket states
so there's a couple things happening
here we're going from this active agent
model through pearl's work
building on bayesian statistics and
markov blankets talked about other
places other people
can have a lot more to say on that
adding
as per mel andrews's paper the
innovation of fristin
which is the partitioning of blanket
states into sensory in action
so sensory as being defined as inbound
and active as being defined as
outbound and then we're gonna take this
sort of like equations of motion
trajectory flow remix

and think about that kind of a physical
depiction of a system
with the partitioning between now two
flows like a
highway with two directions so here was
just like
landscape flow like the elevation is is
a flow in one way
it's kind of just its own flow the water
flow but then
here we're gonna have two directions in
a way it's not just up and down hill but
it's
like there's a sense flow and an action
flow so there's information coming in
that might have an asymmetry and then
there might be like a totally disjoint
kind of asymmetry with respect to action
going out
so if it has a robot has a photon sensor
and then it has a
physical actuator this is going to allow
a total obvious really simple
partitioning yeah photon come in
actuator go out generative model
internal
maybe i know that fully like a state
machine
maybe i'm modeling it like the
intentional stance but by partitioning
into
sensory input and then the motor the
behavioral experimental output
we're going to be able to kind of like
tidy our
thought around interfacing with active
systems and even inactive systems
and then formally that's defined as

um being a flow so that's this f
is going to be a function of policy
which is like the agent's
you know it is the policy of your arm to
be the length it is just like it's the
policy of someone to wear the clothes
that they wear
and then it's going to be over um
there's going to be this is not from the
same paper but formally there are ways
to define these different states and
partition
them and it's an area of active research
and discussion which formalisms apply
over which
scope and how well all of these
innovations of first ones line up
with all the other thoughts in the area
anything on
markov blankets or this kind of
partitioning
so something i've been thinking about
like while you've been talking and just
a metaphor that keeps popping up for me
is like a
a ski resort right like with the input
information coming in and like depending
on your policy you might select like a
black
diamond or a blue diamond or bunny hill
like slope
and the flow of people going out and
then there's like the people that like a
high level of surprisal that like want
to ski back country and like possibly
hit a tree or
like there's all of this like you know
all of these options um that's kind of

how it's been coming together in my
brain yep i the ski slope people have
different affordances
different preferences different risk
tolerances different um inclination for
nostalgia of a simple one versus the
thrill of a new one it's like
these are all it's kind of like what
ryan smith has been talked about like
these are
above the level of just one repeated
task that
two people could perform and oh there's
a difference in how they performed
we're looking at something that's
actually internal about those higher
order properties
all right here is where we go from the
flow state
across an interface so we have this
bi-directional freeway
um with the flow that's partitioned so
we got like signal processing dynamics
on the inbound and all the bayesian
statistics
that kind of interfaces with all the
cybernetics and the policy selection
and then output is behavior so that's
the bi-directional flow
across an interface we can
um use okay let's read this quote and
then
unpack these um equations and then read
them
if the probability distribution q of mu
q sub mu is sufficiently similar to the
actual conditional
distribution p of uh i think it might be

eta
but i i don't know it looks like an n
yeah curly n given
pi policy so p of of um the actual
we'll just call it actual yeah actual
fancy one so if the probability
distribution you know q
of fancy n is sufficiently similar to
the actual conditional
distribution p of n given policy so
states given policy how things will turn
out
given how i act over external states
how will the external states be given my
investing strategy
what will stimuli will i see on my
website
in under different policy assumptions um
then we can approximate that flow so if
you're treading water
some motor pattern is being enacted
that's resulting in treading water
it's not accidental so that is
self-organization
in a dissipating environment and so it's
as if
a system that's treading water is able
to approximate the flow of the actual
states informationally it can stay above
water
informationally so it has to be doing
something like swimming
equivalently we can say that the
gradient flow of autonomous states can
be considered as a gradient flow on
variational
free energy so now we move
from the sort of general computational

phrasing or physical
computational phrasing of gradient flows
into gradient flows being related to
this variational free energy estimate
which it turns out is going to allow um
very
easy like tractable optimization based
upon
minimizing variational free energy so
the
phrasing of a gradient flow was sort of
neutral with respect to what you did
with it
now by porting the gradient flow into
gradients of variational free energy it
opens up a domain of optimization which
is based upon variational
bayes and variational free energy and
just to skip to this
bottom line the variational free energy
so this thing
that it looks as if it's optimizing with
respect to policy
big f of pi um
uh is just equal and it throws out
approximations
coefficients it's just like you gotta
work through it and there's probably a
lot of ways to do it but
um this free energy value like
on policy is related to
this expectation of
this fancy j which are we do we have
that in a later slide
um of basically yeah it's the it is um
the self surprise i think we have a
later slide so
um it's the expectation of how surprised

you are
given about your policies given the
states
so that's like how surprised are you
about where you are
that's why it's about yourself surprise
about policy given state estimates
that value plus
the divergence which is like a kl
divergence
between the state estimate
versus the actual flow and there's
probably a lot to say about
these kinds of information theoretic
divergences
um but it turns out that that is going
to be
greater than or equal to how surprised
you are on
policy itself so almost reading it
backwards it's like how surprised you
are to find yourself doing a policy
which is not just an instantaneous
action a policy is the trajectory
like i can't believe i'm the kind of
person who's on the top of mount everest
that is if you've been like you know a
700 pound person down to your couch at
one point in your life like i can't
believe i'm on top of mount everest
would be kind of like i've taken a
policy that i've ran every day and hiked
all this stuff and now
i'm not at first yeah so yeah so we
what we really want to know is given a
um optimistic self identity
a persisting self identity now some
people get hung up on that they're like

what if my self identity is negative
well it'd be equivalent to um the heart
or lungs
stopping existing it'd be then you
wouldn't see that
system treading water and that's why
it's
tragic when it does happen so we want to
think about
how surprised are individuals with
respect to policy this doesn't tell you
how to know what to expect or what
policies are possible but this is
how surprised one is about their own
policy
is going to be approximated with an
expectation so it's tractable
of surprisal of policy given states
which is much easier to calculate
much easier to say how surprised am i
about um how much i've been investing
given how much i have
rather than computing the total space of
everything that could happen which is
infinite plus just the details
of how far off you are so if you're on
the ball
then this divergence term is low and
then
your the variational free energy is like
kind of closer
to the idealized minimal surprise and
this is something that one does want
minimal surprise on because
in a dissipating world you get the
disorder for free
and so this is something that's actually
being like asymptotically converged

towards
so something i um maybe we should maybe
clarify
in the paper i think that the authors
use like self-information
or surprisal they don't they don't
really talk about self-supprisal
um and this is kind of like we're
conflating now with like last week's
paper about um like
who is doing who is creating the model
of the self like when
you have a self model who is exactly
constructing that self model and i think
like in terms of this it's like
surprisal over policy okay but like
in relation to your model of yourself or
i think your model of yourself is part
of like your generative model
overall yeah maybe yeah interesting
cool let's go to um 29
all right so this was so after thinking
about
this landscape of flows being
operationalized as a variational free
energy
landscape that we're going to be able to
do some kinds of operations
on um now
there can be sort of dimension play
with mapping states from this is
something i would
like to ask the authors about is like
which connecting which manifolds to
which manifolds matter what's the hookup
we need
which ones are the fep providing us with
because in this quote they connect it to

the information geometry
and geometry is just the shape of things
so it's kind of like
the shape of that information landscape
you can think about the
physical landscape or you can say the
flow
on that landscape of the water if the
water fell on it
and so it's kind of like okay
information geometry is one thing
and then this is partitioning with an
information geometry
angle according to this
markov blanket active inference
partitioning
so the quote says and then blue you can
give any thought on it
conversely on this extrinsic manifold so
viewed from the outside the internal
states minimize variational free energy
when conditioned on the blanket states
in other words whenever the blanket
states change that could be sense or
action
the expected internal state changes and
there is movement on the
extrinsic manifold that by construction
so is this how things are or how we're
constructing them
can be cast as bayesian belief updating
this is because the
expected internal state encodes
probabilistic
i.e bayesian beliefs about external
states they act as if
there's an internal state that's
encoding or representing or at least

acting as if it encoded or represented
actions about external states the
blanket states here can be construed
as the sensory impressions of the
external states on the market blank
blanket that contain the internal states
in short it may be more useful to
consider the system from the point of
view of the extrinsic information
geometry
the probabilities encoded by internal
states like neural firing patterns
going out rather than from the point of
view of the
intrinsic information geometry in other
words
the statistical complexity of the neural
dynamics
considered alone might be misleading we
might want to actually
highlight the neural dynamics
projected onto a manifold of action or
with respect to external states
because we don't just want that raw time
series like
doing bitcoin analysis we want to know
how that is associated with other
patterns
in a way that would suggest that it's
acting purely responsibly or proactively
or
in some way that we're going to talk
about how you can describe the
complexity of it
i don't really have anything to add i
thought but i just was looking back
through the paper i had thought that it
was

um you know mapping from one the
manifold of the internal
geometry or like there was a geometry
created that that
was over the whole internal states
versus the external states and it was
that mapping that
that mattered but but uh yeah i'm
curious to revisit that now
so um we have a question in the chat
and someone asks can you say a few words
about the islands of awareness
so that's like the with the sensory
input is disconnected
can you say a few words about islands of
awareness and the relationship to the
equations you're explaining
so on this markov blanket model on
27 if the sensory input
were damped that would be the islands of
awareness model
so if nothing else active inference has
a process theory
that is consistent with the free energy
principle is helping us
functionally operationally however you
want to think about it get at this
island of awareness problem
so instead of just saying imagine that
there was very low or minimal input or
no input of sensory state or
imagine that somebody could get sensory
state but couldn't act
or imagine the only sense they could get
was hearing
and then the only action they could have
is this
this is a way that we can think about

totally separating
the sort of north and southbound traffic
with the active and the sensory states
being partitioned
that's the fristen innovation and so
the island of awareness was from the
non-fep
people it's probably a from a different
area
and so to um kind of latch onto that key
topic that's being more broadly
discussed
this paper is saying actually like
here's the immediate value add of fep
and active inference
we can actually subsume the debate
around islands of awareness
which is something you all already
agreed was an important question
and then we actually have our own way of
thinking about island's awareness
so maybe if that's a question you care
about maybe
um this is a good way that you would
want to study it because
it led us to these interesting
conclusions about islands of awareness
um okay so
30. um okay so this is
unpacking a little bit more about that
big f
of pi with the expectation of the
surprise of
policy given states plus the divergence
of the actual
and the um estimated states so we can
look at the quotation
here the divergence between the

estimated and
actual states describes the statistical
complexity of the internal model
how much do i have to change my mind to
accommodate current changes in sensory
signals
so with respect to state estimates about
how
bright it is it's like
about whether you estimate it to be
bright and it is or it isn't
um whereas this other term the negative
expectation
of how surprised um their one is on the
policy given states
describes the accuracy and then this is
a parenthetical describing why it's
negative
so minimizing this entails minimizing
the difference between these two
so it's accuracy and
model complexity and then it's just
you depending on how you have the
negatives and the positives
you get good points for being accurate
but it's a monotonic penalty to be more
complex
so always better to fit better always
better to be simpler
and those two lines always intersect
because you can always have a simpler
model that does worse
and you can always have a more complex
model that does better if only because
it explains some
little random feature of the error
signal so this is not about how the
world is

this is about statistical partitionings
like a principal component analysis
the principal components are guaranteed
to be orthogonal well
there's certain assumptions that get
built into here that we don't have to
over interpret with respect to how
reality actually
is note that the accuracy depends on the
particular states so the states the
particle
the pi the total states now actually
confusingly
i wonder if this is referring to blanket
and internal states
instead of policy as usually we use pi
to mean
i had that same question oh yeah yeah
during written on my paper yeah because
yeah that's our little icon with the
policy pie but this one's cool too i
like the blanket in internal states but
yeah maybe it could be clarified and
which one is
in here because are we thinking about
the minimization on policy alone or
um on the entire thing so let's make
sure we're
clear on that one during a period of
partial disconnection from blanket
states
so such as dreaming that's the islands
of awareness
the variational free energy gradient in
the following equation will mainly be
driven by the complexity part which does
not depend on the blanket states
so we have the bi-directional traffic

and the flow
and then we're disconnecting a sense
so now it's like the system is just
emitting
so to speak flow but that is being
driven
by the um uh divergence term
rather than this including the internal
state policy
we'll figure it out but this part drops
away so then
um because it's not conditioned on these
um blanket states it's only conditioned
on the internal model and actual model
so this d term is only conditioned on
internal actual states and this one also
includes
blanket states so
you can then do some
good factorization or mathematics and
then the conclusion of that
at least um or the form of this equation
entails
that neural processes can perform the
computations required
to minimize free energy even when the
coupling with the environment is
suspended
it's like you unplug the computer or
something and then it's like it stops
but then
could you seize sensory input for the
brain in a way that actually
um let the brain reveal
a different stream of output that would
reflect any complex internal dynamics
now how will we measure it we'll get to
in a second but the idea would be that

we could use sleep and dreaming as a
model
i don't know and think about how the fep
could be applied to these islands of
awareness
and we come to this same result isolated
systems will minimize free energy by
reducing the complexity of the
generative model
blue i knew i knew you thought that was
interesting what did you like about that
um so i i um specifically liked uh
the fact that they said like in when
you're sleeping
there's no accuracy right like or like
in a in some island of awareness
type of state if you don't have any
accuracy because you don't have any
um sensory input and so when in these
situations that then
you minimize free energy by reducing the
complexity of the generative model
i think that that's cool because um
sleep like just from a neuroscience
like perspective background like it's
really thought to like integrate and
solidify
like your experiences shuffle things
from short-term memory into long-term
memory if like they're
you know necessary like if it was like a
peak emotional experience it'll be like
cemented forever
but all these things happen like during
the sleep phase so it's literally
like reducing complexity of the neural
correlative consciousness also which i
think is interesting

yep very nicely said like when you're
getting
sensory data in there's an imperative to
make it match
like to predict what the next word is so
that your world doesn't seem like this
like blur
of just things that are uncertain but
then when you disconnect the sense
the only imperative the only reason that
was getting complexity pushed up was the
need to fit data
but when you don't need to fit the data
there can be this radical scaling back
in the complexity of the bottle
so it's kind of like play or learning or
higher dimensional
existence or lower dimensional pretty
interesting okay
so 31 this is going to be a connection
now from that variational free energy
landscape
to computational research in
optimization and machine learning
they're going to reveal variational free
energy minimization in machine learning
was predicated on
minimizing algorithmic or computational
complexity
which is the basis of universal
computation
people can read the citation and have
their own thoughts on what it means to
be universal or be a computer but
there's the citation
here's pretty interesting part
consciousness researchers assume
maybe uncited who does maybe but

one could assume complexity has to be
large computational complexity has to be
large yet the imperatives for universal
computation in general
and the free energy principle in
particular say exactly the opposite
so are we looking for at the simple
rules end or at the complex outcomes end
this apparent paradox can be resolved
easily
by noting that free energy is abound on
marginal likelihood
aka model evidence in the same way that
the lz which we're going to get to in a
second complexity is an upper bound on
algorithmic complexity so they're like
it's not enough to just look at how
surprised you are by individual neural
states because the brain is never going
to repeat
an exact state so you're going to be
infinitely surprised if you'd use like
parametric statistics you need to
actually have a type of
statistical inference based upon the
dynamics of the trajectories
and interpreting them on a flow
landscape of information geometry
rather than just being infinitely
surprised by the specific measurements
you get
and then they basically
write that to kind of speak to the
island's awareness idea
it's no surprise then to see that a good
model of a complex world
will show a high degree of algorithmic
or statistical complexity even when

observed in temporary isolation
so this is saying okay you unplug the
computer it's going to radically
simplify
if it was a bitcoin node it's
disconnected from the internet it's
going to simplify its processing
but there might be another kind of
system that has a
high degree take your empirical question
what high degree or
shouldn't they be simpler or wouldn't it
simplify once it's isolated
but that's the kind of interesting
behavior we would expect or be
predicting to see
from an isolated system and just my
question for the author was
what does it mean that the free energy
is bounding the model evidence
in the same way that this information
measure which we're going to get to of
complexity is an upper bound on
algorithmic complexity
which is very curious it's a lot of
things i've only heard a little bit
about
so it's just very interesting to think
about what did this make you think about
um well so i think i put what i thought
about this in a few slides
away from now we'll we'll loop back into
that i think
cool so then we're like okay what is
this lz complexity because blue and i
both love complexity and like met while
we were studying complexity
a few years ago so we were both excited

with that
and this levels if complexity measure
was
our statistic however you want to think
about it was first presented in the
article
on the complexity of finite sequences in
1976 by two israeli computer
scientists uh lempel and ziv this
complexity measure is related
to the kolmogorov complexity but the
only and that's so then another one
would be the shannon so shannon is like
the surprise with the coin flip
kolmogorov is like the algorithmic
complexity usually described as that way
but the only function that the lz
is going to use is the recursive copy ie
the shallow copy
the lz complexity can be used to measure
the repetitiveness of binary sequences
and text like
song lyrics or prose fractal dimension
estimates of real world data have also
been shown to correlate with lz
complexity
so it's interesting because learning
about it on the wiki page
this is apparently the whole algorithm
it's kind of just like
looking for repeating blocks
blocks with prefixes and pointers
it's a compression technique so if it
was something where if your file was one
two three four five
you'd imagine that it would take
i don't even want to say too much
because i don't even know what would

have a high or low lc complexity i'd
like to ask
but just like something could be easily
zipped or not
some things are going to be a minimal
amenable to lz complexity measures being
high
or low so that was just my main question
is how do we go from this algorithm to
like
conscious systems or to measuring real
comparing real
systems just pretty interesting not sure
what to think about that but also really
looking forward to the um
everyone's perspective in like 16.1 and
two because it was out of
i didn't expect this to come into play
it's cool to learn about what do you
think about it
so i looked at i looked up like the um
lz complexity versus like the kolmogorov
complexity where the komodo or
complexity is like total randomness
right like 3.14159 like that's like a
totally like common goal of complexity
to the max
but this lz complexity i think is
related to like
repetitiveness and sequences and how
much they're repeated
so it's a it's different than the just
totally
randomness i i guess that was like my
glint of what i i read from that so yeah
if something was just
10 of the same version you know
transposon or something concatenated 10

times
you could define it by defining the unit
and then 10 pointers
you wouldn't need 100 data points you'd
need 10 plus 10 pointers so like for
example that would be something
compressible
um related to lz but let's take a step
back and ask
okay even beyond the measure of um
the complexity or the entropy measure i
mean
let's learn and find out what all those
are but
what is dynamical complexity and this is
really related to complex adaptive
systems and a lot of the more
like non-free energy principle
complexity research that we've been
learning about blue
so what is it rather than how it can be
measured now we're not going to fall
into a realist trap we're not going to
say systems are this way but
we're thinking about the kinds of
complexity that could be dynamically
instantiated and measured and this is a
paper from
um first in 2019 a not b which was the
particular physics one
and in dynamical systems theory so this
is drawing from even before
complexity in some ways um like systems
theory
three types of complex behavior can be
distinguished so this is not
feedback loops and stuff like that those
are the mechanisms

by which we're gonna be tracking one
particle in the space so we've got the
particle
the one ping-pong ball in the airflow
that's the particle trajectory model
and then we have the flow which is the
air that the ping pong ball is being
spun around in
in that sort of vector flow trajectory
world
but sometimes we see chaotic itinerancy
so that's like um when you have two
pieces of wood in the stream and if it's
a low the opponent experiment they go
together for a long time if there's a
lot of chaoticness
then high chaotic divergence they
diverge rapidly
heteroclinic cycling so this is like
cycling
with um so repeating to come back to
some places but
often with a twist and then switching
under multi-stability so like you can be
like your brain can be like in a sleep
mode
and then it can go into the waking mode
and so yes there is
a gray zone but you tend to see um like
phasic or categorical transitions so
complex systems
do that kind of stuff sometimes they're
wild and chaotic itinerancy and wacky
and then they can also cycle and also
they can switch and sometimes we can
even
un couple those ideas like something
could switch and the switching could

reflect
it turning on or off chaotic itinerancy
or something like that
so and then i just copied out the
definitions from the chapter
in the first book what do you think
about that or what did that make you
think about with respect to complexity
just like the dynamical complexity
versus like you know what's
happening in a computer is like
um just how things change really over
time right like so
in in a dynamical system is a system
that's changing not
like how complex is this number that
is pi or you know 1.000 right so it's
just that that basic difference
cool um okay let's return to
the kind of free energy perspective
again thinking about the dynamical
complexity but we want to be thinking
about
the complexity of these dynamical
systems um
so this is because the computation is
performed by a system that minimizes
variational free energy and inferences
which correspond to movements on the
systems extrinsic statistical manifold
existing measures of dynamical
complexity however do not measure
complexity in terms of probabilities
encoded by neural activity
but in terms of statistical properties
of neural activity
that is properties of the intrinsic
statistical manifold hence dynamical

complexity
at least measured by existing approaches
should be associated with nccs
at least in the point of view the fep so
this is to their point
that we want to move from thinking about
neural states to neural trajectories and
flows
is otherwise we don't get access to
um this kind of a dynamical complexity
measure
because if we just think in terms of
stop snapshot states
we're going to think shannon entropy how
surprised am i by this neuron being
lit instead of unlit at this time slice
whereas
in the flow model we can think about the
dynamics
of trajectories through really high
dimensional state spaces
and that's where fep suggests an
alternative an fep inspired measure of
dynamical complexity
would be based on the information length
of neurally encoded probability
distributions
ie bayesian beliefs instead of
considering the statistical properties
of a neural time series and such so
that's kind of what i just mentioned
all right conscious activity and
complexity blue what would you say about
this
so this goes back to the lz complexity
really
and and also the common goal of
complexity but but i'll just read this

quote from the paper so it says to the
extent that proposed measures of
dynamical complexity
have empirical validity as measure of
consciousness
which is like what is consciousness what
are we measuring
like how do we just know if it's valid i
don't know all those things like
are packed into that sentence for me
they may regard it be regarded as
empirically adequate
but it would still be desirable to
provide a fundamental justification
because this could further support their
application to controversial cases like
islands of awareness
in addition it could help resolve an
uncertainty about whether existing
measures of dynamic complexity also
track
algorithmic complexity so we we know
that lz complexity is
is not um it's not the same as common
goal of complexity because it just takes
in
the repeats but it doesn't take in all
of the different ways that
um pomodoro complexity does so
here it says it's well known that lg
complexity for instance constitutes an
upper bound on
algorithmic complexity however this
entertains the possibility that
conscious activity
maximizes lz complexity and related
measures of dynamical complexity but it
minimizes algorithmic complexity

this hypothesis is stated and there's
the citation ruffini 2017.
finally developing a measure of
dynamical complexity within the fep
could help providing uh help by
providing a computational explanation of
consciousness
so i think that that's a really cool um
thing that the authors have brought back
around like can we
can we put dynamic complexity in the fep
framework
um and they're thereby you know put a
computational explanation of
consciousness forward i think that's
really awesome
great points okay let's talk a little
bit more about a formalism that
people love and also just an interesting
um
i think question about realism and
instrumentalism and also just how we
think about
the definitions and which things are
contingent on which so
here they're going to write a further
assumption we shall make is that the
system possesses a markov blanket so
the language of whether it possesses or
is
or there just is one and then the system
is a secondary description
and it describes a little bit more about
the blanket and so this was kind of
related to earlier but we didn't need
this
level of depth and it talks about where
it's drawn on

in terms of developing on perl and the
directed
acyclic graphs and generalizations from
there
but then here what matters is that the
existence of a markup blanket has subtle
profound implications so is that the
markup blanket exists
in the world or is it that the
implications of us
modeling with a markov blanket have
profound implications
so how do we think about whether we're
describing how the systems
are versus being clear um when we think
that it's actually
not how things are or where they might
be
congruent and so is it how do we know
and this is related to a general
question people has how do we pull out
these markov blankets
um from the empirical data
and relate it to the analytical concept
so again another slide just from mel
andrews's paper
uh discussion where we talked about the
development of the markov blanket topic
from markov's initial framing in
analytical mathematics and matrix
mathematics to pearls extension
into bayesian statistics and graph
theory computational approaches
and then where fristen has developed it
over the decades
and collaborators and then what is now
being um done and learned about
with respect to this the post markov

blanket whatever it's wanting to be
called
any markov thoughts
yep no cool um
this is related to one of the global
challenges that were addressed which is
that
mapping where does consciousness come
into play it's like whoa okay
all of this flow and the information
theory and the geometry but
how are we going to connect this back to
the consciousness question and relating
anything about the system to the actual
um question of what is conscious or not
and so this is the one quote that i want
to highlight on 39
is being conscious versus just
simulating it and so
they um write a system with a
non-equilibrium steady state
and a markov blanket comprises internal
external and blanket states
as shown in section 3 such a system can
also be described as performing certain
computations
inferences by virtue of causal relations
between internal external and blanket
states
that is what we were measuring those
relation dynamical contingencies
in the equations we talked about but in
by virtue of the causal relations this
is different for a mere simulation
to be sure a simulating system can
perform the same computations
and it will perform them because of
causal relations between its parts so

like it is true that the a matrix and
the b
matrix get multiplied together in the
program
however it will not perform them because
of causal relations between its internal
external blanket states
the simulation may instantiate a virtual
machine
which may have internal external blanket
states such that its internal states
encode or probability distribution over
external states given blanket states
but the internal states of the virtual
machine will not
be the internal states of the physical
machine on which
the virtual machine is running so this
is getting into the total
virtualization simulation everything and
we're not going to read this dialogue
but on 40
it's a section from a
conversation about this topic in the
book the minds eye which which is um
compiled by
hofstetter and dennit and just it's
about
you're simulating a hurricane and oh
well programmers aren't claiming the
simulation as a hurricane
it's just simulating parts of a
hurricane buildings don't get wet
and then you have to look at it the
right way well what do you mean look at
it the right way you know what is
getting wet
it's your perception it just it's it's a

big area so it's fun because
it's something that you know we're all
just learning about and
again no one has a great measuring
device so it's not like we can
resolve these debates there's not like
an open and shut answer so
that's what's so interesting is like
there might be a hot take or a quote you
like or something that's inspiring to
you about whether computers are
conscious or not
but in the end it's almost like
it's a bottomless pit so that's why it's
good that the authors are actually
starting to build bridges across it i
think
all right and then a last um one
slide on reservoir computing this was
suggested by sarah
she was bringing up reservoir computing
and researching
maybe things that we'll hear about in
the coming months like about how
it might relate to active inference
reservoir computing is a framework
for a computation that is sort of
generalizing or building upon a neural
network
model and it's mapping the inputs not
just to
a fixed array of nodes or even multiple
layers like a deep neural network or
something
but the way that these nodes internal to
the system the reservoir
they can actually be any system
so that could be various types of things

it could be a slowly changing or fast
changing system or system with a lot of
other structures
and so just we we don't have answers to
this but bringing up just
reservoir computing sarah had some
thoughts on it so we'll look forward to
hearing about this
and about the niche as a reservoir and
computation as
um happening and where does that put
computationalism
any thoughts on that before we go to our
last bit nice
chill and these were just some things we
brought up but
you know people will bring up their own
in 16.1
uh and anyone else can bring up uh
anything they want us to put on the
screen for the author and others
but there's so much to think about with
this paper
they concluded just by writing that
um how can research on neural correlates
of consciousness lead to an explanation
of consciousness and what role could
free energy principle
play in this endeavor we have suggested
that research on neural correlates
should be complemented by research on
computational correlates of
consciousness
in order to yield a computational
explanation of consciousness moreover
according to free energy principle
neural correlates to consciousness must
be defined in terms of neural dynamics

not states and cccs should be defined in
terms of probabilities encoded by the
neural states
so specific claims for those to engage
with
a lot of um footholds and
challenging walls to climb but
interesting topics
a lot of things that understanding could
enable
and we're talking about consciousness
and neuroscience so
everything is kind of adjacent to it any
closing
thoughts blue on this very nice
discussion
no thank you so much for having me it's
been fun participating
yep that was great times thanks um so
much for
joining because it makes it a lot you
know more interesting to make and to do
so um we hope to see people in 16.1
and beyond other than that